Thin Film Logbook

User manual

October 22, 2026

Abstract

The Thin Film Logbook is a web application for recording what happens to every sample in the lab: where it came from, each process step (deposition, annealing, implantation, ...), each analysis (XRD, RBS, SEM, ...), and the photos and data files that go with them. This manual explains how to use it day to day. Section 10 covers the extra tasks for administrators, and Section 11 how to run and back up the server.

Contents

1 Getting started **3**

1.1 Opening the logbook . . . 3

1.2 Your account . . . 3

1.3 Finding your way around . . . 3

2 Key ideas **4**

2.1 Sample IDs . . . 4

2.2 Who can see what . . . 4

3 Samples **5**

3.1 Creating a sample . . . 5

3.2 The sample page . . . 5

4 Recording steps and analyses **5**

4.1 The form . . . 5

4.2 Entering parameters . . . 6

4.3 Available types . . . 6

4.4 One run on several samples . . . 6

4.5 Photos and files . . . 7

4.6 The timeline order . . . 7

5 Cutting a sample into pieces **7**

6 Sharing and ownership **8**

6.1 Sharing a sample 8

6.2 Performed by 8

6.3 Giving a sample to someone else 8

7 Substrates and projects **8**

7.1 Substrate library 9

7.2 Projects 9

8 Search and export **9**

9 Suggestions **9**

10 For administrators **9**

10.1 Parameter schemas 9

11 Running the server **10**

11.1 What to back up 10

1 Getting started

1.1 Opening the logbook

The logbook runs on a lab computer and is opened in any web browser, on a computer or a phone, at the address the administrator gives you, for example `http://10.10.13.172:8000`. You need to be on the lab network.

1.2 Your account

There is no self-registration: an administrator creates your account. You receive an email with your username and a temporary password (or the administrator gives them to you).

1. Sign in with the username and temporary password.
2. You are asked to choose a new password.
3. You are asked for your **Name** (shown as the researcher on your samples) and your **Initials**: 2–4 letters, unique in the lab. Your initials go into your sample IDs, e.g. `ASS-2026-001`.

Careful

Choose your initials carefully. Once you have samples, only an administrator can change them, and existing sample IDs never change.

If you forget your password, use **Forgot password?** on the sign-in page: you receive an email with a one-time sign-in link. You can change your password and name at any time from your profile (click your name at the bottom of the sidebar).

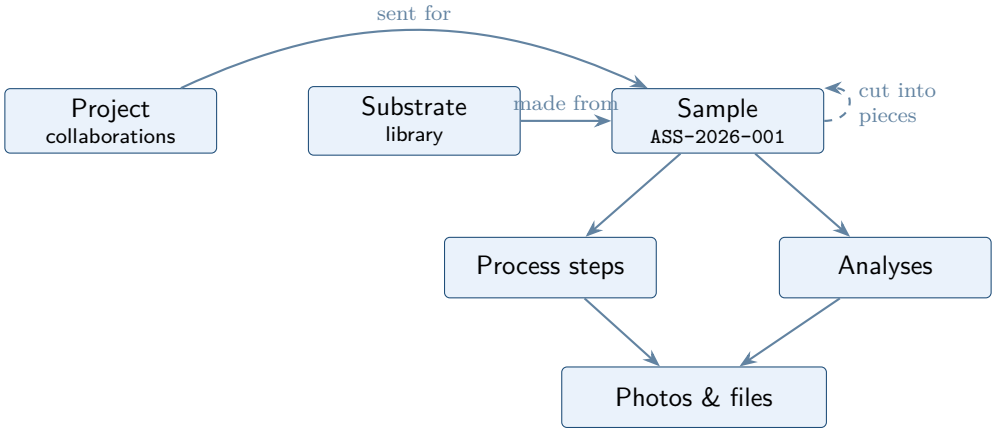
1.3 Finding your way around

The sidebar on the left (a menu button on phones) has:

Samples	All samples in the lab, newest first.
Search	Filter, group, share and export samples.
Substrates	The shared library of substrates samples are made from.
Projects	Collaboration projects with other institutions.
New Sample	Create a sample.
Suggestions	Send an idea, a problem or a question to the administrators.
Inbox, Configurations	Administrators only (Section 10).

At the bottom of the sidebar are the **PT / EN** language switch and the moon/sun button for dark mode. Technical terms (technique names, units, dropdown values) stay in English in both languages.

2 Key ideas



Sample One physical piece of material, with a generated ID that never changes, an owner (the researcher), and a *timeline*.

Timeline The ordered list of process steps and analyses done to the sample, each with its parameters, notes, date, photos and files.

Substrate An entry in the shared library (material, orientation, thickness, polishing, supplier, spec sheet). Thin-film and bulk samples are made from one.

Project A collaboration with another university or centre that sends us samples. Its short code replaces the initials in its samples' IDs.

Pieces Cutting a sample creates new samples that inherit its history up to the cut.

2.1 Sample IDs

`ASS-2026-001`	Researcher ASS, year 2026, their first sample that year
`ASS-2026-001-A`	Piece A cut from it
`ASS-2026-001-A-2`	Piece 2 cut from piece A
`ASS-2026-001-REF`	Reference piece
`UMIN1-2026-001`	First 2026 sample of project UMIN1

IDs are assigned when the sample is saved and never change, even if the sample is given to someone else. Use the optional **Label** for your own description ("waveguide test, batch 2"); it doesn't have to be unique.

2.2 Who can see what

Everyone signed in sees every sample's *summary*: ID, label, type, materials, substrate, tags, the techniques used and the researcher. The *details* (descriptions, parameters, notes, dates, files) are visible only to:

- the sample's owner;
- administrators;
- people the owner has shared it with (**View** or **Edit**);

- for a single step or analysis, the people listed as having **Performed by** it.

Samples without an owner are open to everyone.

3 Samples

3.1 Creating a sample

Click New Sample and choose the **Sample type**:

Type	What to fill in
Thin film	The substrate from the library (required), dimensions, and the film material once it's deposited.
Bulk	The library substrate it's a piece of (required), dimensions, and an **External ID** if someone else cut it.
Commercial	Description, supplier, product/batch number and the spec sheet (PDF or photo, required).
Collaboration	The project (required), a description (required) and, optionally, the partner's own ID for the sample.

All types take a **Label** and **Tags**. If the substrate or project you need isn't in the list, follow the **Missing one?** link: it opens the form in a new tab; add it there, then reload the sample form.

The **Researcher** field defaults to you. Pick someone else to create the sample for them (Section 6.3).

3.2 The sample page

A sample's page shows its summary at the top, then the **Sharing** panel (for the owner), then the timeline. From here you can:

- **Edit** the sample, or **Delete** it (owner and admins only; not while it has pieces);
- **Add Process Step** and **Add Analysis**;
- **Cut into pieces** (Section 5);
- reorder the timeline (Section 4.6).

4 Recording steps and analyses

4.1 The form

Choose the type first; the form then shows that type's fields.

Date Optional, the day only. The **Today** button fills it in.

Where CTN, INESC MN or another institute.

Equipment The machine used. The list only offers equipment at the place chosen under **Where**. (On a deposition it's called **Machine**.)

Performed by You by default. Add colleagues, or **Someone else** to type the name of someone without an account.

Parameters The type's own fields (below).

Other parameters Anything else, one `key: value` per line.

Notes Free text.

Photos and files Can be added here or later from the timeline.

4.2 Entering parameters

- **Numbers** have their own box and unit. Pressures are in mTorr and can be typed as `1e-4`; a decimal comma also works.
- **Durations** take `min:s` (`12:30`), `h:min:s`, or plain minutes (`12`).
- **Dropdowns** offer the values saved earlier. Choose **+ Add new...** to type a new one; it is then offered to everyone. Please check the list first, so the same target or gas isn't spelled three ways.
- **Variants**: some types ask first which variant it is, e.g. Annealing → Furnace or RTA, and then show that variant's fields.
- **Modes** (XRD / HRXRD): tick each mode measured ($2\theta/\omega$, rocking curve, grazing incidence, ...) and fill in its range, step, acquisition time and configuration.
- **Tick boxes** reveal extra fields, e.g. **Calibration sample** on a deposition.

4.3 Available types

Process steps	Deposition, Annealing (Furnace / RTA), Ion Implantation, Lithography (optical / e-beam), Etching (dry / wet), Cut, Other Process
Analyses	XRD / HRXRD, XPS, SEM, TEM, RBS (RBS / Channeling), AFM, Raman, Optical Microscopy, Hall / 4-probe, Ellipsometry, Profilometry, Other Analysis

Administrators can add types or change their fields (Section 10).

4.4 One run on several samples

When the same deposition or measurement session covers several samples, record it once: under **Also add to other samples**, tick the other samples. Each gets its own copy of the entry, linked together as a run ("Also on: ..." on the sample page).

- Photos and files in the main fields go to every sample (stored once).
- A file field appears next to each ticked sample for that sample's own data.
- You can add more samples to a run later, from the edit form.
- When editing one copy, you can apply the changes to the others. If that would overwrite or clear a value another sample already has, you're shown which and asked first.
- Each copy keeps its own place in its sample's timeline, and deleting a copy removes it from that sample only.

4.5 Photos and files

Each card on the timeline has **Add photos** and **Attach files**; you can also drag files onto a card. Uploading starts as soon as files are chosen.

Add photos On a phone this opens the camera or gallery; pick several at once. Large photos are rotated upright and reduced to at most 2560 px / 2 MB JPEG; iPhone HEIC photos are converted. Location data is removed.

Attach files Stored exactly as uploaded. Use this for raw data (e.g. 16-bit TIFFs from SEM or AFM, `.raw`, `.csv`), so nothing is lost.

Images get a thumbnail on the timeline; click it to open the full file.

Tip

Photograph the sample, the sample holder or the screen of the instrument straight from your phone at the bench: open the sample's page, tap **Add photos** on the step's card, and take the picture.

4.6 The timeline order

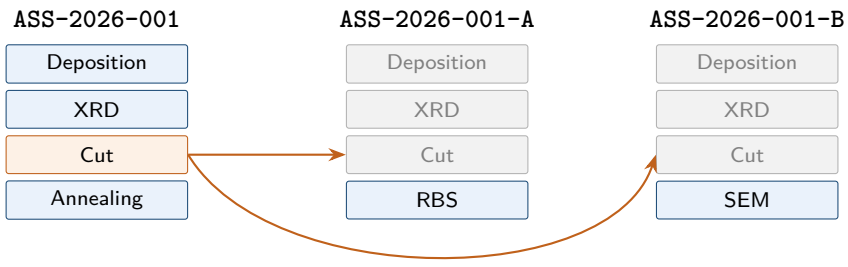
Steps and analyses share one timeline. A new dated entry is placed after everything dated on or before that day; an undated entry goes at the end.

- Use the up/down arrows to move an entry, e.g. to order two things done on the same day.
- **Sort by date** puts the dated entries back in date order, keeping your order within each day. Undated entries stay where they are.

5 Cutting a sample into pieces

Cut into pieces on the sample page records a *Cut* step and creates one new sample per piece. It's next to **Add Process Step** and **Add Analysis**, and it's the only way to record a cut.

1. Choose who did the cutting under **Cut by** (you, by default).
2. Choose the **Number of pieces**. Tick **Also keep a reference piece** to add one more (`-REF`).
3. Take a photo of the pieces with each piece's letter written next to it.
4. For each piece, optionally give a label and dimensions, and choose its researcher (by default, whoever cut it).



grey = inherited, read-only on the piece

Each piece:

- **inherits** the parent's timeline up to the Cut step, shown greyed out with "from ...". Nothing is copied: correcting an entry on the parent corrects it for every piece. Inherited entries are edited on the parent.
- copies the parent's type, substrate, film, tags and sharing.
- gets its own entries after the cut, separate from its siblings.

The parent stays usable; what you add to it afterwards belongs to the parent only. Whoever can see a piece also sees its inherited history.

6 Sharing and ownership

6.1 Sharing a sample

On a sample's page, in the **Sharing** panel, choose **Share with...**, pick a person and **View** or **Edit**:

View Read the details and download files.

Edit Also add and change steps, analyses and files.

To share many samples at once, go to Search, tick the samples (or **select all mine**) and use **Share selected samples with**. Only the owner or an administrator can change sharing or delete a sample.

6.2 Performed by

Listing someone under **Performed by** lets them see and edit that one entry (and its photos and files) even without access to the rest of the sample. They are told by email. This is the easiest way to let the technician who ran the XRD upload the data.

6.3 Giving a sample to someone else

You can make another researcher the owner:

- on the New Sample form (the ID then uses their initials);
- on the sample's **Edit** form;
- per piece on the cut form.

Unless an administrator does it, the new owner has to confirm. They get an email and see the sample under *Assigned to you — please confirm* on the Samples page, where they can **Accept** or **Decline** (which gives it back). Until then, and afterwards, whoever assigned it keeps edit access.

7 Substrates and projects

7.1 Substrate library

The Substrates tab lists substrates with material, thickness, orientation, polishing, size, supplier, notes and spec files. Anyone can add one; whoever added it and administrators can edit it. A substrate used by a sample can't be deleted.

7.2 Projects

The Projects tab lists collaborations: name, partner institution, code, country, contact and notes. The code (2–6 letters or digits) is used in the IDs of the project's samples, so it can't match anyone's initials, and it is fixed once a sample uses it. An institution can have several projects; pick it from the suggestions so it's always spelled the same way.

8 Search and export

Search filters samples by free text, researcher, sample type, material, technique and tag, and groups the results (by researcher, year, type, material, technique, tag or project). Free text matches IDs, labels, descriptions, notes, parameters and project names, but only on samples whose details you can see.

Export results (CSV) downloads the matching samples, with one row per step or analysis and one `param:...` column per parameter. It opens in Excel, Origin or Python (`pandas.read_csv`). Details are left blank for samples you can't view.

9 Suggestions

The Suggestions tab is for anything: a bug, a missing field, a new dropdown value you can't add, an idea. It records which page you were on. Administrators reply there; you'll see a badge when they do, and you can answer back.

10 For administrators

Configurations has three parts.

Users Create accounts (a temporary password is emailed, or shown on screen if email isn't set up), reset passwords, make or remove administrators, change someone's name or initials, delete users.

Process Step Types / Analysis Types Add, edit or delete (when unused) types. Each has a *parameter schema*: a JSON list of fields. Editing it never changes stored data.

Export full database (CSV) Everything, regardless of sharing.

The Inbox collects everyone's suggestions; reply, and mark them solved or reopen them.

10.1 Parameter schemas

Each field has a `key`, a `label` and a `type`:

type	entered as	extra keys
`number`	a number box	`unit`, `placeholder`
`duration`	`min:s`, `h:min:s` or minutes	`unit`, `placeholder`
`text`	a one-line box	`placeholder`
`longtext`	a large box	`placeholder`
`choice`	a shared dropdown	`list`; `located:` `true` for equipment
`variant`	pick one option, each with its own fields	`options`
`check`	a tick box revealing fields	`fields`
`modes`	tick several; each gets the same fields	`list`, `fields`

Example (Annealing):

```
[{"key": "method", "label": "Method", "type": "variant", "options": [
   {"value": "Furnace", "fields": [
     {"key": "temperature", "label": "Temperature", "type": "number", "unit": "°C"},
     {"key": "duration", "label": "Duration", "type": "number", "unit": "min"}]},
   {"value": "RTA", "fields": [
     {"key": "atmosphere", "label": "Atmosphere", "type": "choice", "list": "atmosphere"},
     {"key": "temperature", "label": "Temperature", "type": "number", "unit": "°C"},
     {"key": "duration", "label": "Duration", "type": "number", "unit": "s"},
     {"key": "ramp", "label": "Ramp", "type": "number", "unit": "°C/s"}]}]}]
```

11 Running the server

The full details are in the repository's `README.md`. In short:

```
cp .env.example .env          # set SECRET_KEY, BASE_URL, SMTP_PASSWORD
docker compose up -d
docker compose exec app python -m backend.create_admin
```

11.1 What to back up

Database (samples, timelines, users)	the `pgdata` Docker volume
Photos and data files	the `uploads/` folder
Settings and passwords	`.env`

```
docker compose exec -T db pg_dump -U logbook logbook > backup_$(date +%Y%m%d).sql
```

Keep the dump together with a copy of `uploads/`. That folder has one subfolder per sample ID and, inside it, one per step or analysis, with the files under their original names (e.g. `ASS-2026-001/2026-10-22_XRD-HRXRD/scan01.raw`), but it's the database that links each file to its entry.

Careful

`docker compose down` stops the logbook and keeps the data. `docker compose down -v` **deletes the database**.